Kernix 0.10.0 / Performance brief

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

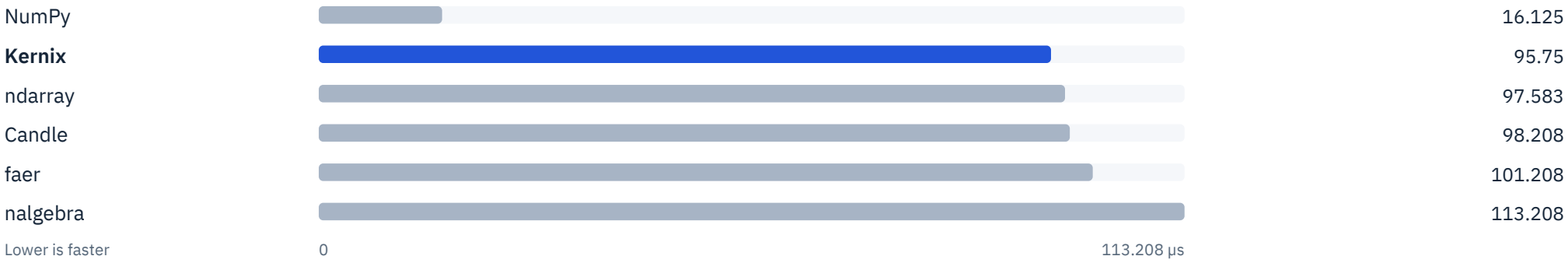
34 workloads

116 observations

Source precision retained

Matrix multiplication

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

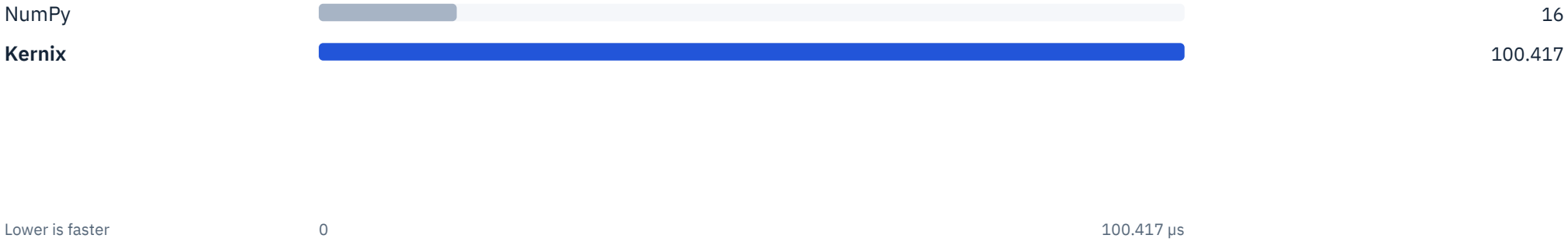
Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Independent matrix operands

Per-call median

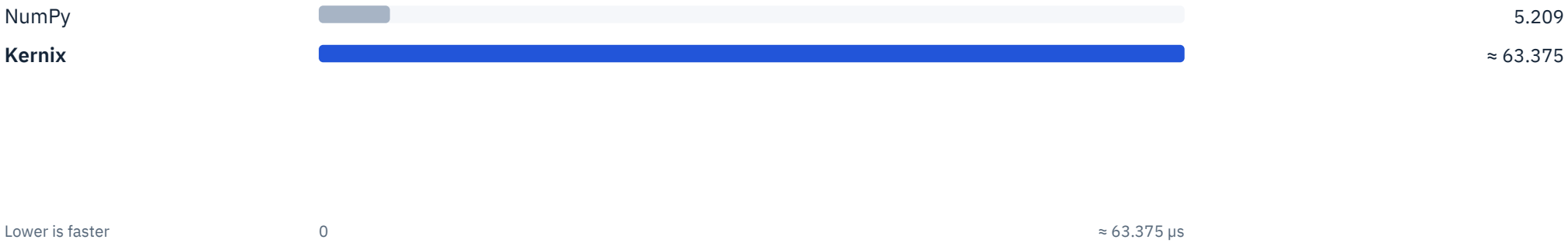
128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · independent operands



Independent matrix operands

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · independent operands



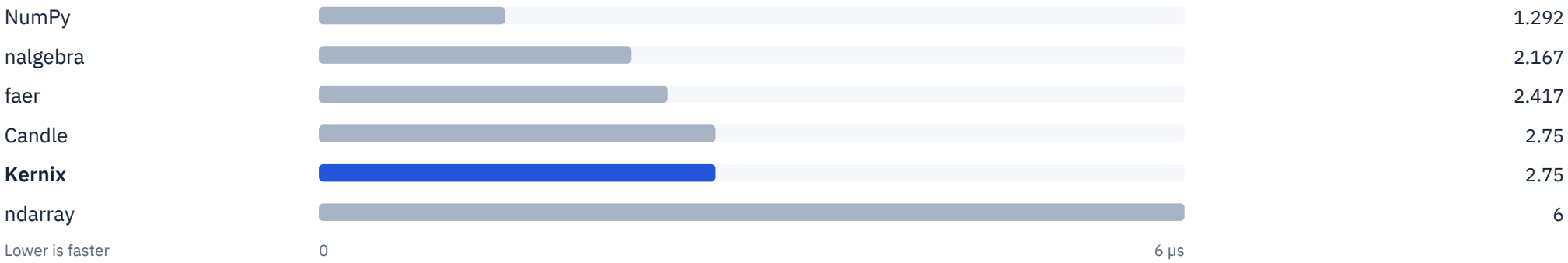
Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

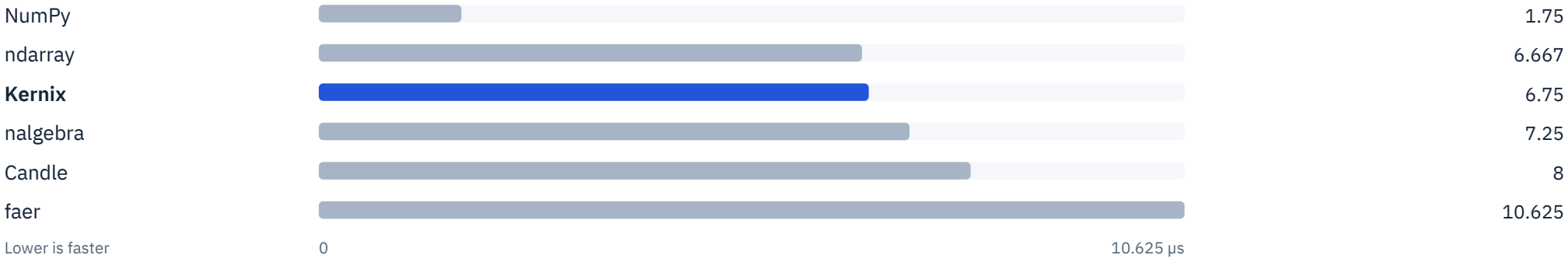
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product



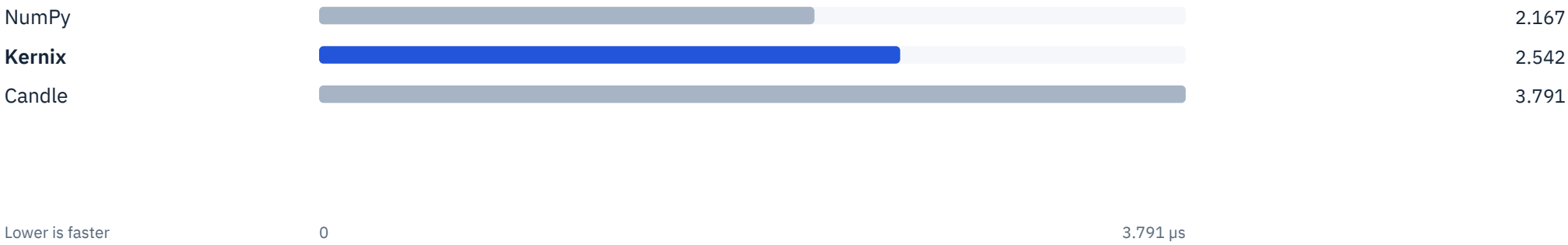
Kernix 0.10.0 / Matrix multiplication

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Batched multiplication

Per-call median

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.10.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

64 × 64 · 64 × 64 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f64 · 2 workers · reused output · pool preparation excluded



Kernix 0.10.0 / Explicit parallel execution

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Matmul · 2 workers · reused output

Per-call median

256 × 256 · 256 × 256 · f64 · 2 workers · reused output · pool preparation excluded



Matmul · 2 workers · reused output

Per-call median

128 × 128 · 128 × 128 · f32 · 2 workers · reused output · pool preparation excluded



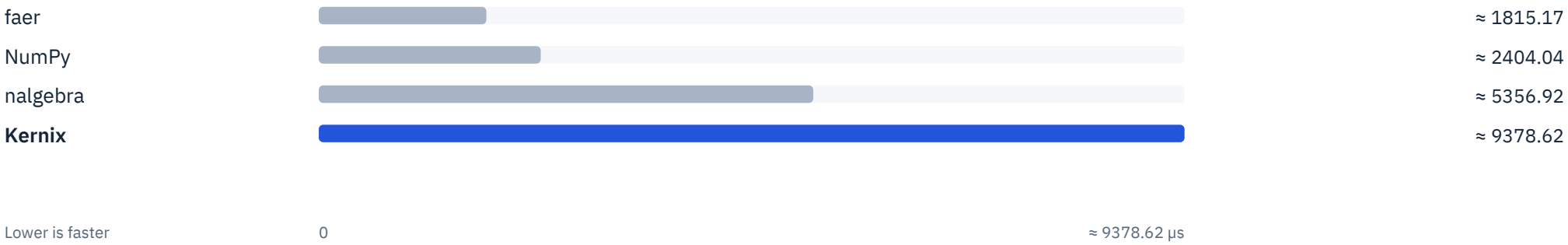
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

512 × 512 · f64 · 1 worker · allocated output



Column-pivoted QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



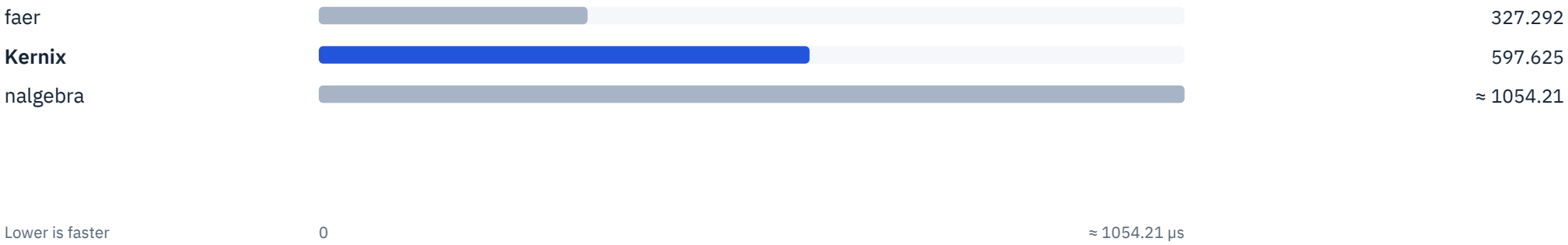
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Column-pivoted QR

256 × 64 · f64 · 1 worker · allocated output

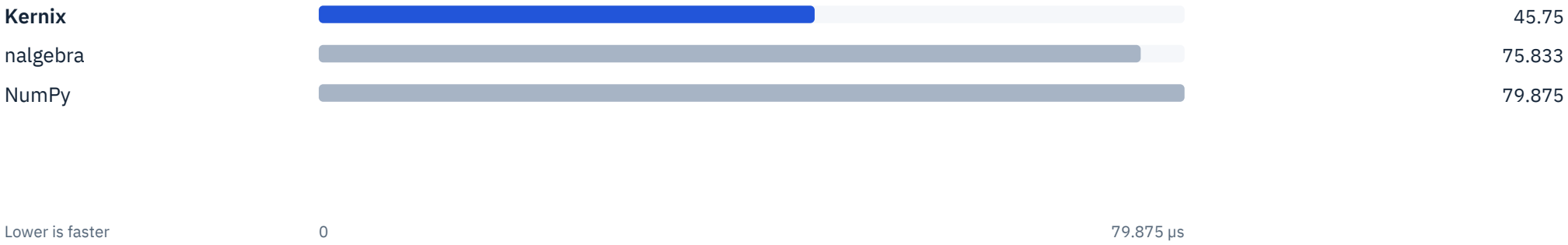
Per-call median



Minimum-norm least squares

64 × 32, rank 16, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ

Per-call median



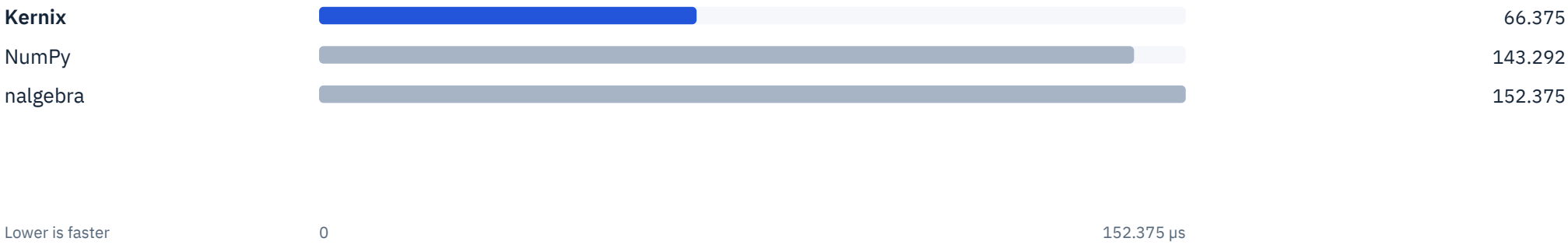
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Minimum-norm least squares

Per-call median

32 × 64, rank 32, 4 RHS · f64 · 1 worker · 4 RHS · factorization included · algorithms differ



Thin QR

Per-call median

96 × 32 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

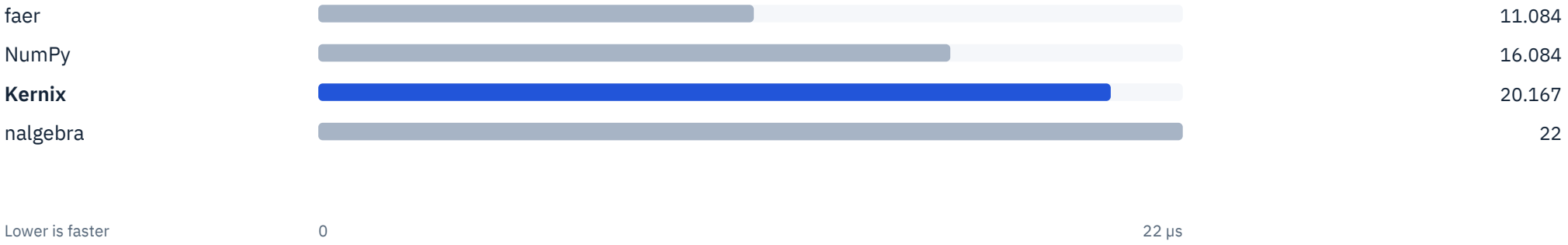
Thin QR

256 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



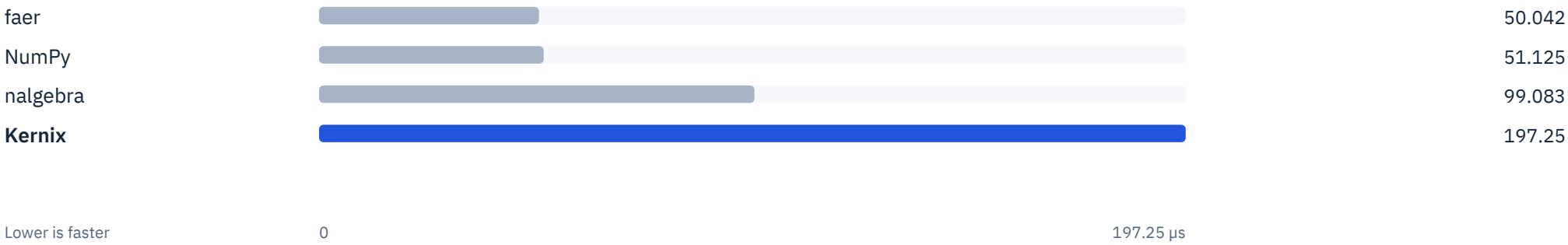
Kernix 0.10.0 / Decompositions & solvers

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Cholesky factorization

Per-call median

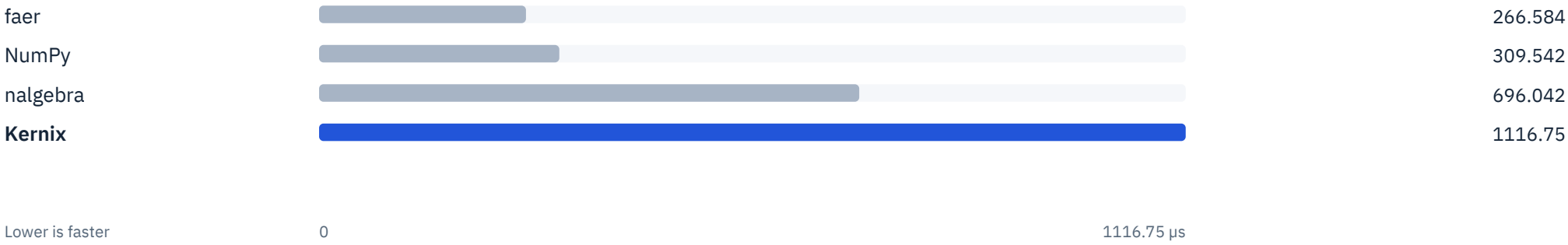
128 × 128 · f64 · 1 worker · allocated output



Cholesky factorization

Per-call median

256 × 256 · f64 · 1 worker · allocated output



Kernix 0.10.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Shared-variable backward

Per-call median

sum(X * X), 64 × 32; prepared graph · f64 · 1 worker · prepared graph · fresh gradients



Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Kernix 0.10.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model recording + forward

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Kernix 0.10.0 / Training & autodiff

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Parameter update

Per-call median

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



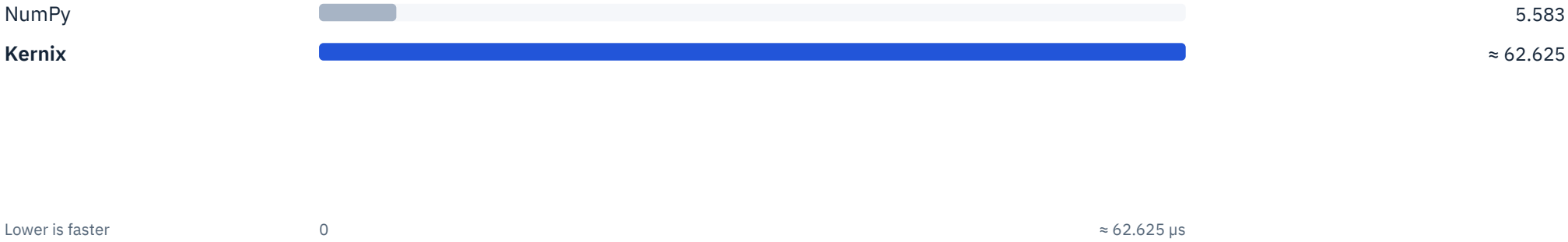
Kernix 0.10.0 / Tensor contractions

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Coalesced tensor contraction

16 × 32 × 8 · 32 × 8 × 16 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.10.0 / Statistics & activations

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

Row softmax

256 × 64 · f64 · 1 worker · allocated output

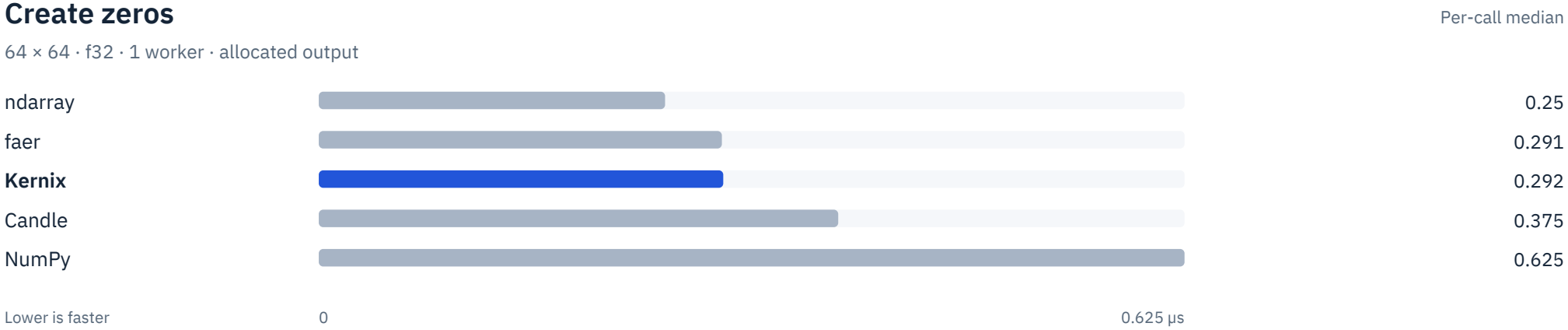


Kernix 0.10.0 / Tensor construction

Apple M1 (Virtual) · Per-call median · µs · NumPy BLAS: accelerate

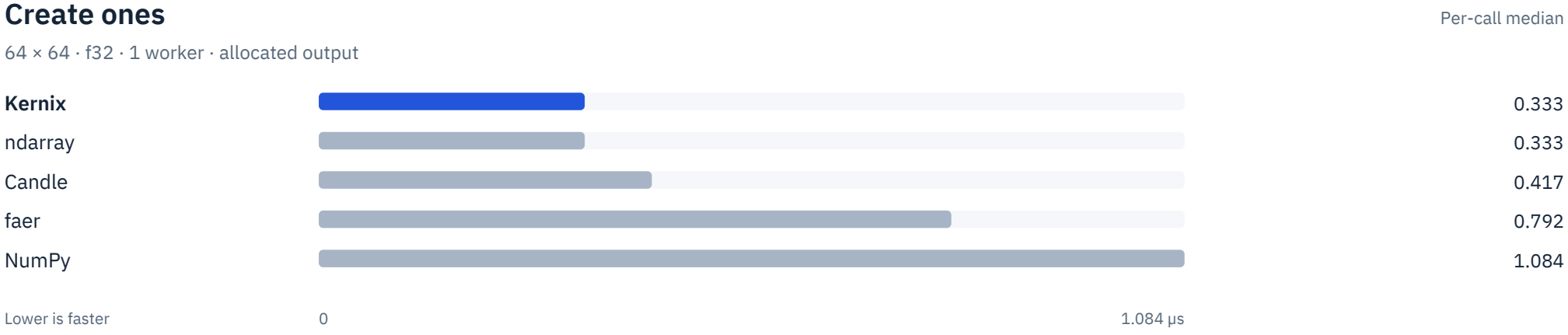
Create zeros

64 × 64 · f32 · 1 worker · allocated output



Create ones

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f64	100417	100.417
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f64	16000	16
Independent matrix operands / NumPy 128 × 128 · 128 × 128 · f32	5209	5.209
Independent matrix operands / Kernix 128 × 128 · 128 × 128 · f32	63374.99999999999	63.37499999999999
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	2750	2.75
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	1292	1.292
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	6000	6
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	2167	2.167
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	2417	2.417
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	2750	2.75

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	13750	13.75
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	13584	13.584
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	13417	13.417
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	12667	12.667
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	3500	3.5
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	14708	14.708
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	16125	16.125
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	98208	98.208
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	113208	113.208
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	101208	101.208

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	97583	97.583
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	95750	95.75
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	854500	854.5
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	123083	123.083
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	772958	772.958
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	853417	853.417
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	790792	790.792
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	887584	887.584
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	7250	7.25
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	8000	8

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	10625	10.625
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	6667	6.667
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	6750	6.75
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	1750	1.75
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	50625	50.625
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	51459	51.459
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	65208	65.208
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	6334	6.334
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	49625	49.625
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	55458	55.458

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	405709	405.709
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	34750	34.75
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	422250	422.25
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	373459	373.459
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	422917	422.917
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	410542	410.542
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	3791	3.791
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	2167	2.167
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	2542	2.542
Matmul · 2 workers · reused output / Kernix 64 × 64 · 64 × 64 · f64	38250	38.25

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f64	141291	141.291
Matmul · 2 workers · reused output / Kernix 256 × 256 · 256 × 256 · f64	454917	454.917
Matmul · 2 workers · reused output / Kernix 128 × 128 · 128 × 128 · f32	52375	52.375
Cholesky factorization / nalgebra 512 × 512 · f64	5356917	5356.917
Cholesky factorization / NumPy 512 × 512 · f64	2404042	2404.042
Cholesky factorization / faer 512 × 512 · f64	1815166	1815.166
Cholesky factorization / Kernix 512 × 512 · f64	9378625	9378.625
Column-pivoted QR / Kernix 96 × 32 · f64	68833	68.833
Column-pivoted QR / faer 96 × 32 · f64	39791	39.791
Column-pivoted QR / nalgebra 96 × 32 · f64	92958	92.958

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Column-pivoted QR / faer 256 × 64 · f64	327292	327.292
Column-pivoted QR / nalgebra 256 × 64 · f64	1054208	1054.208
Column-pivoted QR / Kernix 256 × 64 · f64	597625	597.625
Minimum-norm least squares / nalgebra 64 × 32, rank 16, 4 RHS · f64	75833	75.833
Minimum-norm least squares / NumPy 64 × 32, rank 16, 4 RHS · f64	79875	79.875
Minimum-norm least squares / Kernix 64 × 32, rank 16, 4 RHS · f64	45750	45.75
Minimum-norm least squares / nalgebra 32 × 64, rank 32, 4 RHS · f64	152375	152.375
Minimum-norm least squares / NumPy 32 × 64, rank 32, 4 RHS · f64	143292	143.292
Minimum-norm least squares / Kernix 32 × 64, rank 32, 4 RHS · f64	66375	66.375
Thin QR / Kernix 96 × 32 · f64	57417	57.417

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 96 × 32 · f64	45041	45.041
Thin QR / NumPy 96 × 32 · f64	52958	52.958
Thin QR / nalgebra 96 × 32 · f64	44917	44.917
Thin QR / faer 256 × 64 · f64	332917	332.917
Thin QR / Kernix 256 × 64 · f64	557541	557.541
Thin QR / NumPy 256 × 64 · f64	416083	416.083
Thin QR / nalgebra 256 × 64 · f64	359708	359.708
Cholesky factorization / NumPy 64 × 64 · f64	16084	16.084
Cholesky factorization / nalgebra 64 × 64 · f64	22000	22
Cholesky factorization / faer 64 × 64 · f64	11084	11.084

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 64 × 64 · f64	20167	20.167
Cholesky factorization / Kernix 128 × 128 · f64	197250	197.25
Cholesky factorization / nalgebra 128 × 128 · f64	99083	99.083
Cholesky factorization / faer 128 × 128 · f64	50042	50.042
Cholesky factorization / NumPy 128 × 128 · f64	51125	51.125
Cholesky factorization / faer 256 × 256 · f64	266584	266.584
Cholesky factorization / Kernix 256 × 256 · f64	1116750	1116.75
Cholesky factorization / nalgebra 256 × 256 · f64	696042	696.042
Cholesky factorization / NumPy 256 × 256 · f64	309542	309.542
Shared-variable backward / Candle sum(X * X), 64 × 32; prepared graph · f64	4000	4

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Shared-variable backward / Kernix sum(X * X), 64 × 32; prepared graph · f64	8000	8
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3292	3.292
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7208	7.208
Model inference / Candle batched normalized/sliced MLP · f64	5708	5.708
Model inference / Kernix batched normalized/sliced MLP · f64	2417	2.417
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3208	3.208
Model recording + forward / Candle batched normalized/sliced MLP · f64	6375	6.375
Model backward / Kernix prepared graph → fresh gradients · f64	3209	3.209
Model backward / Candle prepared graph → fresh gradients · f64	18958	18.958
Parameter update / Kernix transactional SGD, fixed gradients · f64	708	0.708

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Coalesced tensor contraction / Kernix 16 × 32 × 8 · 32 × 8 × 16 · f64	62625.00000000001	62.62500000000001
Coalesced tensor contraction / NumPy 16 × 32 × 8 · 32 × 8 × 16 · f64	5583	5.583
Row softmax / Candle 256 × 64 · f64	76250	76.25
Row softmax / ndarray 256 × 64 · f64	65208	65.208
Row softmax / Kernix 256 × 64 · f64	85958	85.958
Row softmax / NumPy 256 × 64 · f64	74416	74.416
Create zeros / Kernix 64 × 64 · f32	292	0.292
Create zeros / faer 64 × 64 · f32	291	0.291
Create zeros / ndarray 64 × 64 · f32	250	0.25
Create zeros / Candle 64 × 64 · f32	375	0.375

Recorded precision / exact latency table

Apple M1 (Virtual) · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / NumPy 64 × 64 · f32	625	0.625
Create ones / Candle 64 × 64 · f32	417	0.417
Create ones / NumPy 64 × 64 · f32	1084	1.084
Create ones / Kernix 64 × 64 · f32	333	0.333
Create ones / ndarray 64 × 64 · f32	333	0.333
Create ones / faer 64 × 64 · f32	792	0.792